Elementary cells, one-bit capacity, and partition invariance

A spectral answer (Doc. 302)

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Doc. 302 Elementary cells, one-bit capacity, and partition invariance — a spectral answer

The following addresses a natural objection to any "quantised space" picture: what observer-independent criterion distinguishes one elementary cell from its neighbours; why a fundamental length should imply fixed cells each storing exactly one bit; whether the resulting entropy is invariant under changes of origin, orientation, coordinates, or foliation; and whether the indivisibility of a cell follows from the Planck scale or is an added assumption. The short answer is that every one of these points is correct *against a spatial-cell ontology*, and that FFGFT does not rest on one. It replaces cells with topological and spectral invariants (winding classes and modes), and the objections dissolve in turn.

.1 The picture the objections target

A "one bit per Planck cell" ontology tiles space (or a horizon area) into fixed cells of size $\sim \ell_P$, assigns each a one-bit capacity, and reads entropy off the count of cells. Against that picture the following points are decisive:

- there is no observer-independent criterion individuating one spatial cell from its neighbour: a tiling carries a chosen origin and orientation;
- the Planck length is a *scale*, and a scale does not by itself imply a partition into discrete containers;
- a partition-based entropy is not invariant under change of origin, orientation, coordinates, or foliation — it inherits the arbitrariness of the partition;
- if space is a continuum, any cell subdivides, so "one cell $=$ one indivisible bit" needs an independent reason why half-cell states are forbidden;
- and absent that reason, the cell is simply an extra postulate riding on top of the Planck scale.

All five are granted. This is precisely why the framework is not built on spatial cells.

.2 Invariants, not cells

The fundamental object is not a lattice of cells but a compact, boundaryless manifold — a four-torus T^4 (with a $\mathbb{Z}_3$ identification), carried into a Hilbert space $\mathscr{H} = L^2(T^4)\otimes\mathbb{C}^3$. The elementary discrete objects of the theory are not regions of space; they are *winding classes* ($\pi_1(T^4) \cong \mathbb{Z}^4$, the carriers of the information unit) and *eigenmodes* of an operator on that manifold (the connection-Laplacian / mass operator, the carriers of the mass spectrum). With that single change of ontology — invariants, not cells — the points resolve in turn.

1. **Observer-independent individuation.** The elementary discrete objects are distinguished by *invariants*, not by a location in space: the information unit by its *winding number* (a topological invariant, $\pi_1(T^4)$), the mass spectrum by *eigenvalues* (a spectral invariant). Neither depends on coordinates, origin, or orientation. There is no "neighbouring cell in space" to distinguish; there are discrete classes labelled by invariants. The individuation is topological or spectral, not spatial — which is exactly what makes it observer-independent.
2. **What the information unit is — and what stays a definition in the "one bit".** The discreteness does not come from the Planck length and is not a tiling; it is *topological*: $\pi_1(T^4) \cong \mathbb{Z}^4$ admits only integer winding numbers. The minimal information unit is therefore *by definition* the winding quantum $\Delta w = 1$ (Doc. 257) — scale-universal and energy-independent; there is no half winding quantum. The identification "one bit ƀ one minimal winding" is a *definition*, though a topologically motivated one (the smallest non-trivial winding), not an arbitrary two-state assumption. The $\ln 2$ is *not* the fundamental unit but its thermodynamic shadow: the *energy* of a bit is scale-specific, $E_{\text{bit}}(L) = \hbar c/L$, and Landauer's $E_L = k_B T \ln 2$ is the special case at a single (the biological) scale. This answers Takayuki's point (2): "why exactly one bit" is a definition on a topological basis ($\Delta w = 1$), not forced by a scale — and the unit is scale-universal, hence precisely not tied to a chosen partition.
3. **Invariance under origin, orientation, coordinates, foliation.** Yes. Because the entropy is a function of the spectrum — a count or trace over eigenvalues — and eigenvalues are invariants, the entropy is automatically independent of origin, orientation, coordinate choice, and basis. This is not an added axiom; it is a property of spectral quantities. A partition-based entropy would fail the invariance test, which is one reason the framework does not define entropy through a partition at all. A further, framework-specific result makes the origin/orientation independence explicit: on the compact structure there is no preferred index assignment — all D_6-equivalent segments of the torus structure are equally valid descriptions of the same object ("meshing, not a cut-out"). No labelling of a "first" cell or a preferred axis is available, or needed.
4. **Continuous space, no half-cells.** Space is continuous: T^4 is a smooth manifold, with no spatial atomism. One therefore cannot obtain indivisibility by subdividing space, and the framework does not try to. The indivisibility is *topological and*

spectral, not spatial: $\pi_1(T^4) \cong \mathbb{Z}^4$ admits no half winding, and the spectrum on the compact manifold has a smallest gap fixed by the topology (the circumference / boundarylessness), just as a finite string cannot vibrate at "half its fundamental". "Half-cell states" is, in this language, a category error: there are no cells to halve. The continuum of the manifold and the discreteness of the spectrum coexist without tension — exactly as in acoustics, where a continuous medium supports a discrete set of normal modes.

5. **Derived, not assumed.** The discreteness follows; it is not an extra "cell postulate". Given a compact manifold, the discreteness of the spectrum is a theorem. What *is* assumed is the compact geometry itself — the torus $T^4/\mathbb{Z}_3$ and the value of ξ. So the honest accounting: the "one-bit-per-cell" postulate is traded for a "compact-geometry" postulate, and from the latter the discreteness is derived rather than injected. That is a real assumption, and it is where the framework should be attacked — but empirically, not by fiat: the same geometry that yields the discrete spectrum also fixes particle masses, and those are falsifiable numbers (for instance a τ mass of 1776.969 MeV, checkable at Belle II). If the geometry is wrong, the spectrum and the masses fail together.

.3 What is assumed, derived, and posited

The geometry itself is also a *description* (Doc. 301, second level) and as such necessarily incomplete; no description reconstructs the object in full. Fundamental in the full sense is only the complete object (third level). That the resonance-faithful geometry traces a *real* object is therefore an explicit ontological *posit* — not a derivation, and not itself falsifiable; what is falsifiable are only its resonance-faithful consequences (the masses). This posit is named as such, not passed off as derived.

Fundamental is the information unit as a *winding quantum* $\Delta w = 1$ (scale-universal, Doc. 257); *energy*, by contrast, is a scale-specific *projection* of the geometry, $E_{\text{bit}}(L) = \hbar c/L$. An *area* shares this projected, scale-dependent character: the four-torus is boundaryless ($\partial T^4 = \emptyset$) and carries no horizon surface for bits to sit on; an area there is a *projection surface*, a second-level object. Where an entropy–area relation $S = N \ln 2$ with $N \propto A/\xi^2$ appears, N counts winding quanta and $\ln 2$ is the thermodynamic shadow; the A is not a fundamental partition of space (which would revive Takayuki's cell worry) but a chosen projection. Fundamental (third level) is neither area nor energy but the complete object; the scale-universal unit on it is $\Delta w = 1$.

The honest accounting is thus: *declared assumption* — the compact geometry ($T^4/\mathbb{Z}_3$, ξ); *definition* — the identification "bit ƀ $\Delta w = 1$", topologically grounded; *derived* — the discreteness ($\pi_1(T^4) \cong \mathbb{Z}^4$), the scale-specific bit energy $E_{\text{bit}} = \hbar c/L$, and the invariance; *empirical exposure* — the masses; *named posit* — the reality of the object.

.4 Summary

The objections are correct against a spatial-cell ontology, and the framework agrees with them by not being one. The discreteness is topological ($\pi_1(T^4) \cong \mathbb{Z}^4$) and spectral (the spectrum on a compact manifold); the minimal information unit is the winding quantum $\Delta w = 1$ (the "bit", by definition, topologically grounded), not the capacity of a spatial cell; and the invariance holds because winding numbers and spectra are invariants, not because a partition was chosen carefully.

.5 Where this is worked out

The discrete spectrum of the T^4 connection-Laplacian is treated in Doc. 283; the Hilbert-space formulation $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$ with masses as eigenvalues in Doc. 282; the information unit as the winding quantum $\Delta w = 1$ and the three-level hierarchy (topological / geometric / thermodynamic) in Doc. 257 (building on Doc. 255); the origin/orientation independence ("meshing, not a cut-out") in Doc. 300. The invariance point in particular deserves a fuller written treatment than it has so far received.